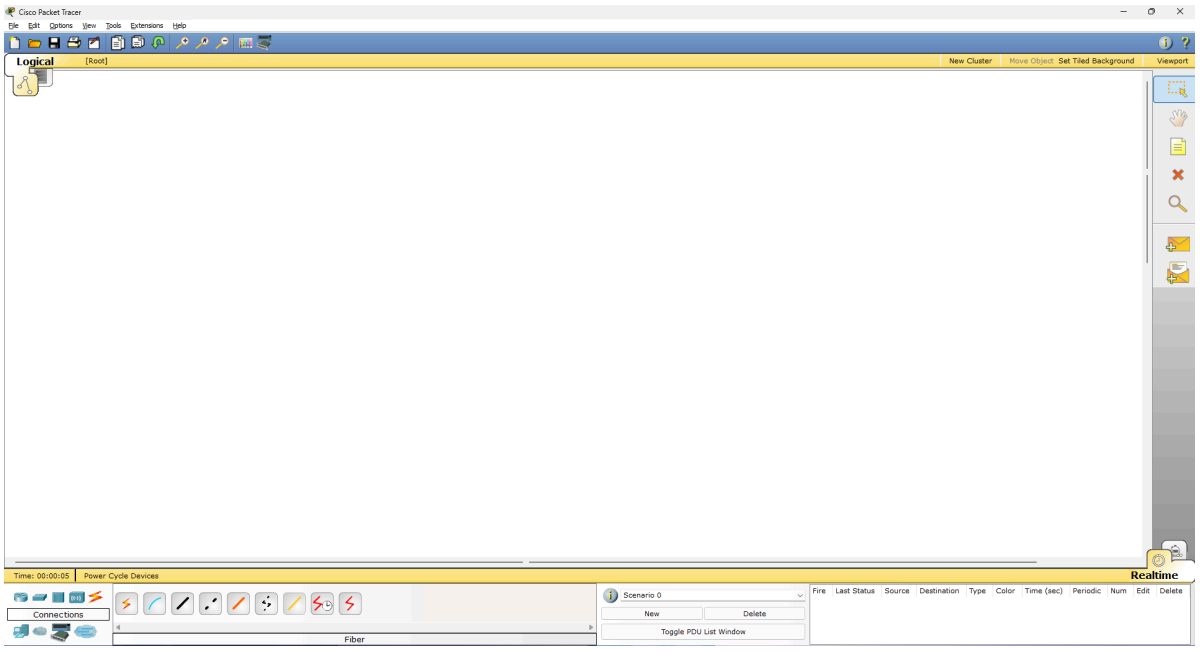
**PRACTICALS**

*Computer Networks Laboratory*

## Practical 5.

**Aim: To implement the topology (atleast two different Network) connected via switch and router in packet tracer simulator.**

Step 1: Open Cisco packet tracer 5.1 as shown in **FIG 1**

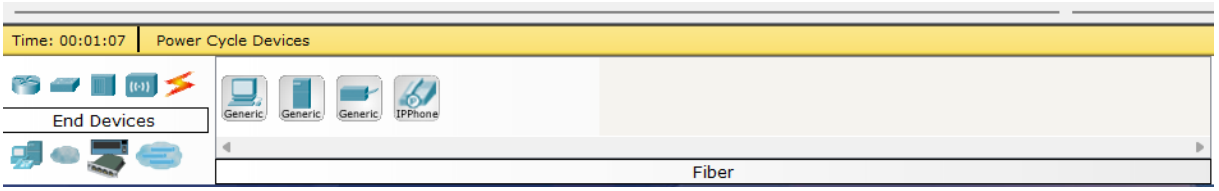



[FIG 1]

Step 2: Select device to make LAN (Local Area Network) Network form the bottom Tool bar as shown in **FIG 2**

There are multiple devices used in this process are follows as:

1. Network Switch
2. Router
3. Hub
4. Wireless device
5. Cables/wires (for connection)
6. End devices (PC, Laptop, Server etc)
7. WAN Enulation
8. Custom made device
9. Multi User connection




[FIG 2]

Step 3: Select PCs, Network Switch, & connecting wires/Cables Shown in **FIG 3**

1. 6 - PCs (3 for each Network)
2. 2 - Network Switch (1 For each Network)
3. 11 - Connecting wires/ cables (5 For Each Network and 1 common For both Network)
4. 2 - Server (1 For Each Network)
5. 2 - Router (1 For Each Network)



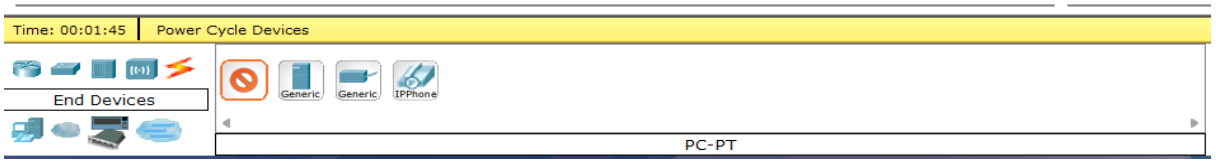




FIG 3

Step 4: Creating Network Using Required devices as shown in **FIG 4**

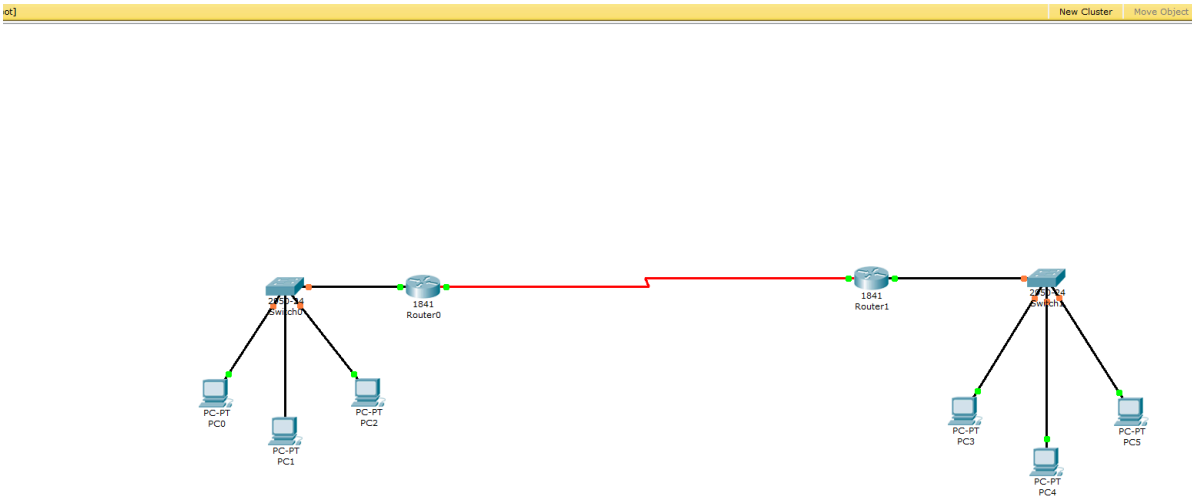



FIG 4

Step 5: Configuring The Setup with Network Using Default Gateway, Subnet Mask, Ip Address as shown in **FIG 5** and **FIG 6**

**Note: This Process Is Only For One Network Do Same Process For All The Networks**

1. Click on PC
2. Go to Desktop Tab
3. Click on Ip Configuration

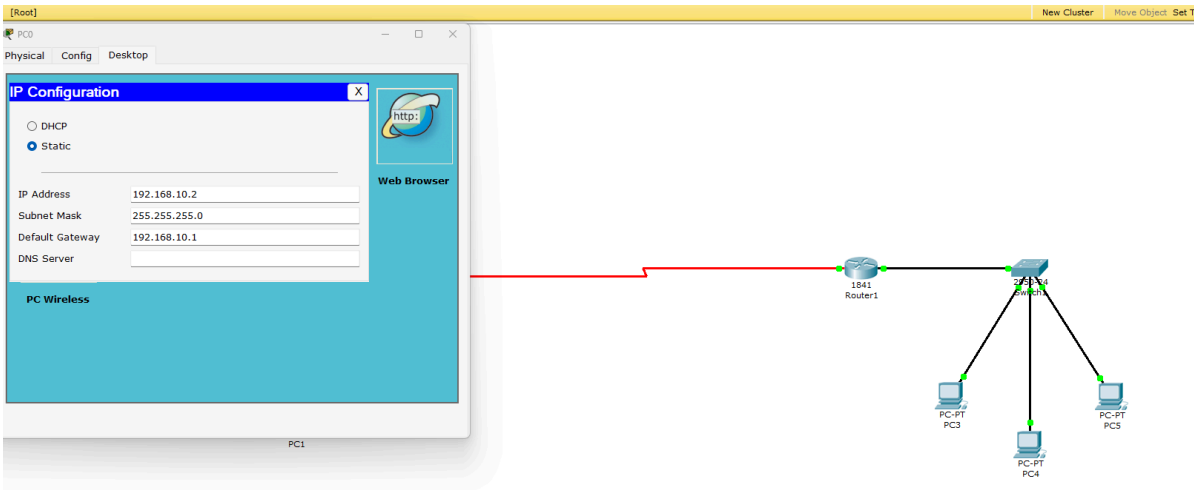



**FIG 5**

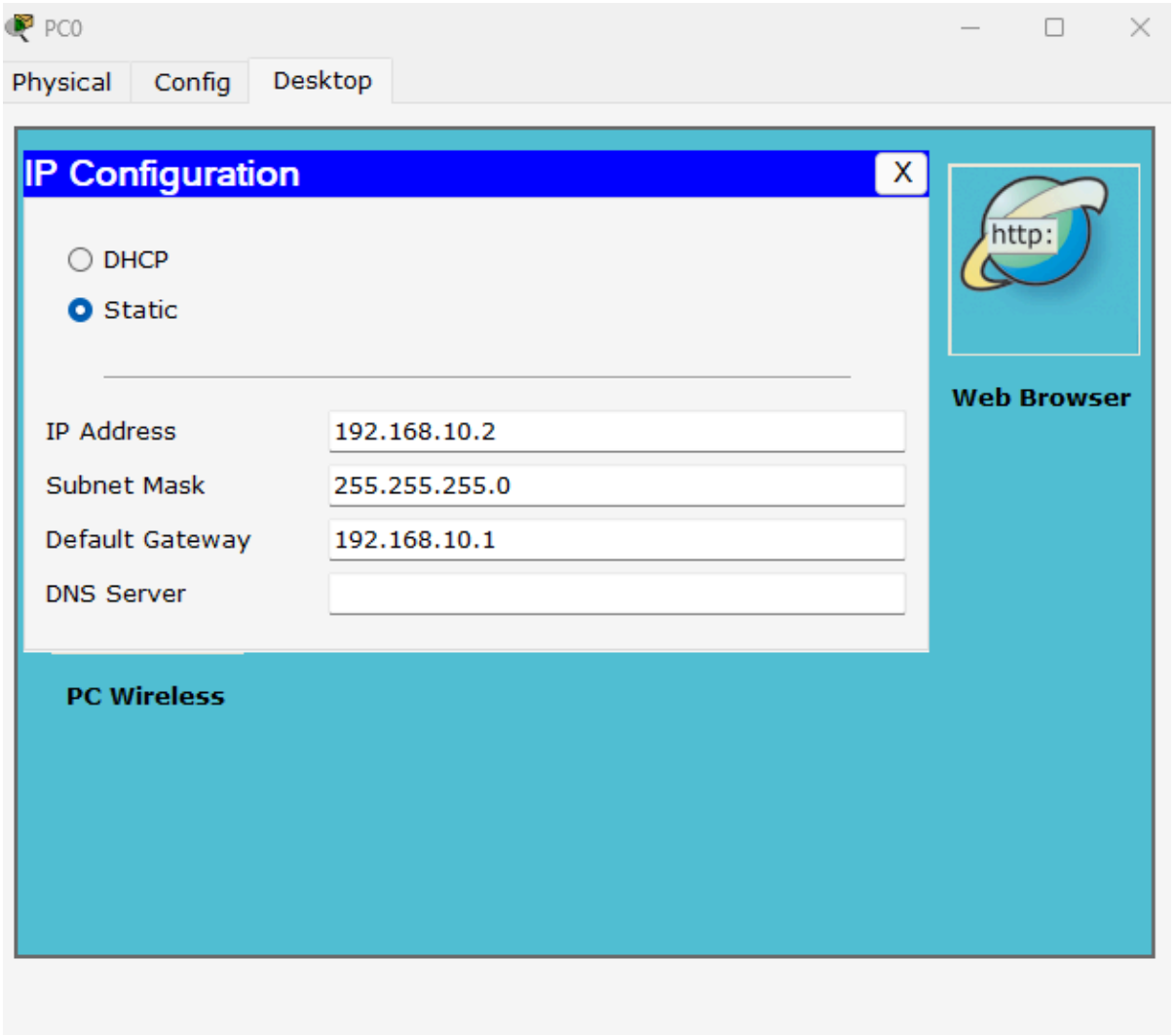



**FIG 6**

**Step 6:** **Now Do the Same Process**

**Step 7:** Now Configuring Routers:

**Router1 Configuration (Network 1 side):**

**Open Router 1**

Add the Following Values in Router 1's Serial 0/0/0 as shown in **FIG 7**

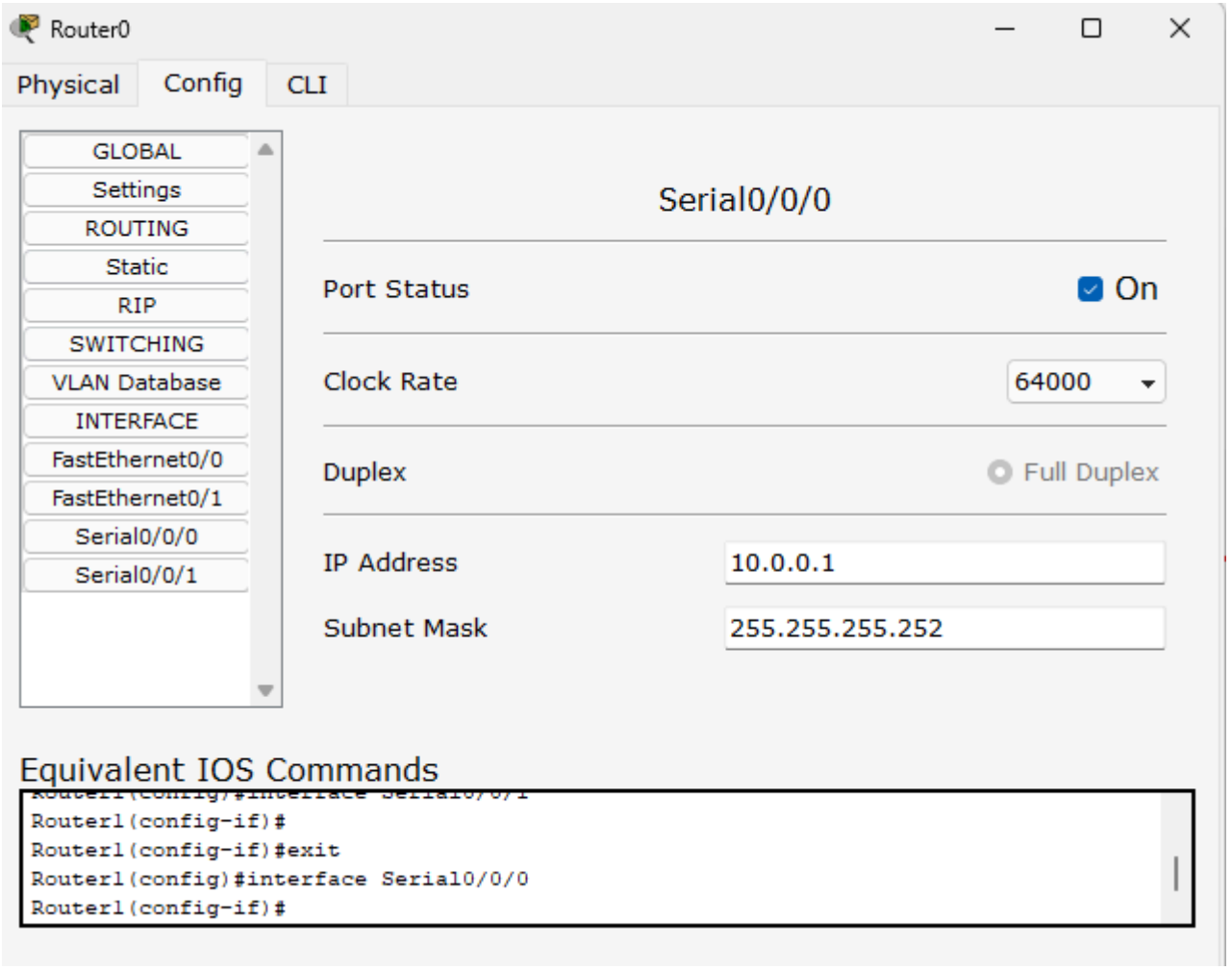

Router0
Physical
Config
CLI
GLOBAL
Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
FastEthernet0/0
FastEthernet0/1
Serial0/0/0
Serial0/0/1
Serial0/0/0
Port Status
On
Clock Rate
64000
Duplex
Full Duplex
IP Address
10.0.0.1
Subnet Mask
255.255.255.252
Equivalent IOS Commands
Router1(config-if)#
Router1(config-if)#exit
Router1(config)#interface Serial0/0/0
Router1(config-if)#

**FIG 7**

Add the following Values in FastEthernet0/0 of Router 1

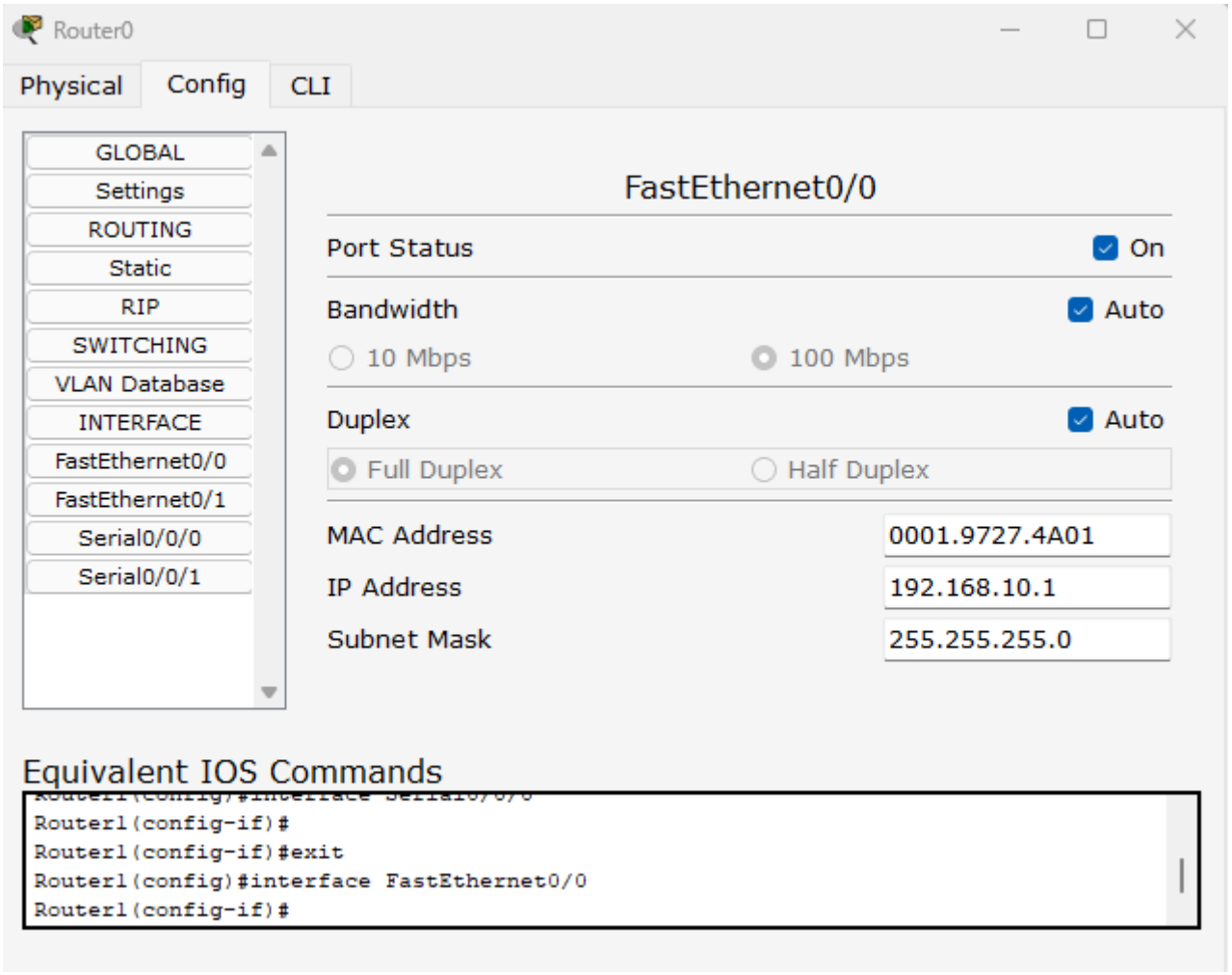



**FIG 8**

**Router2 Configuration (Network 2 side):**
**Open Router 2**

Add the Following Values in Router 2's Serial 0/0/0 as shown in **FIG 9**

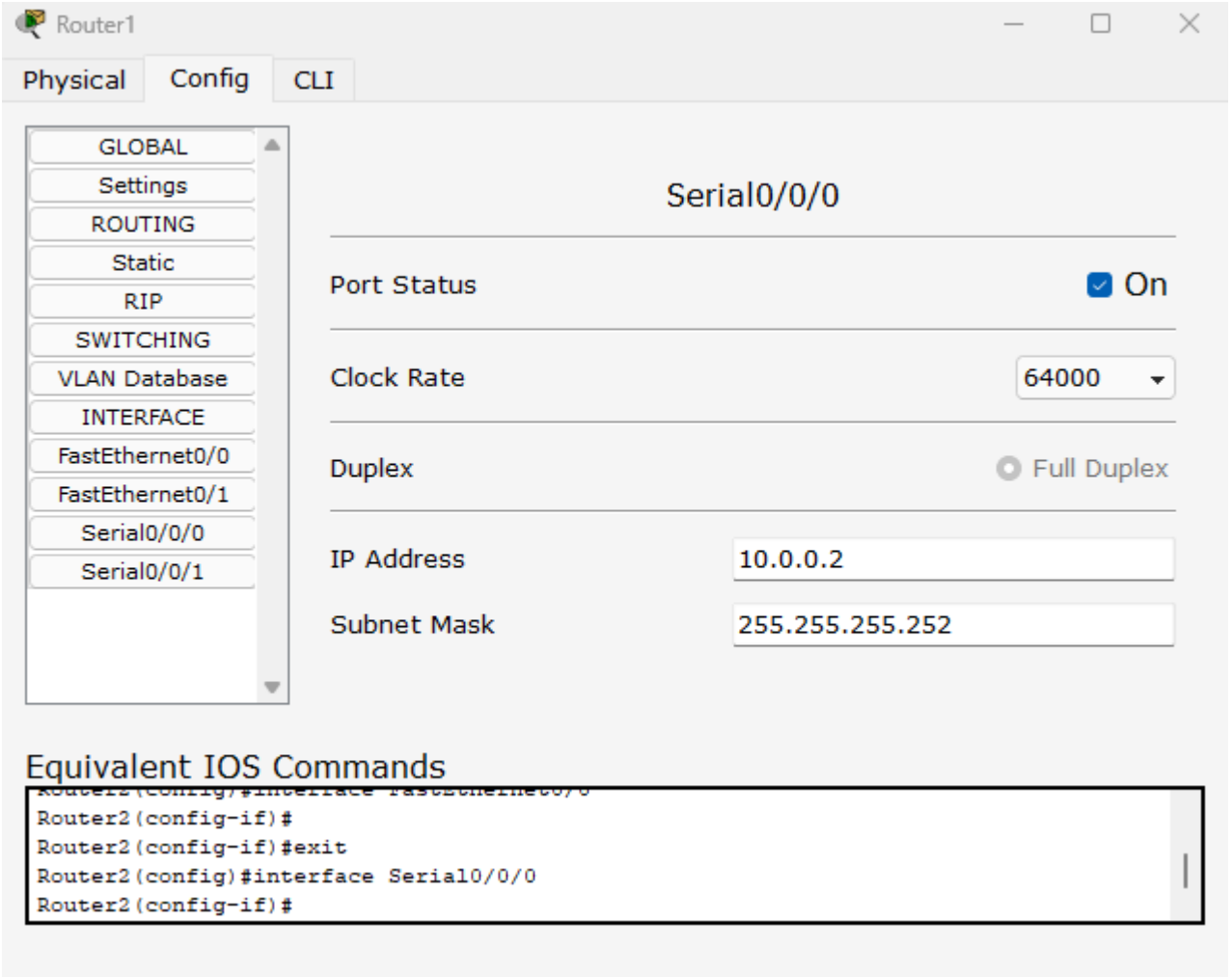



**FIG 9**

Add the following Values in FastEthernet0/0 of Router 1 as Show in **FIG 10**

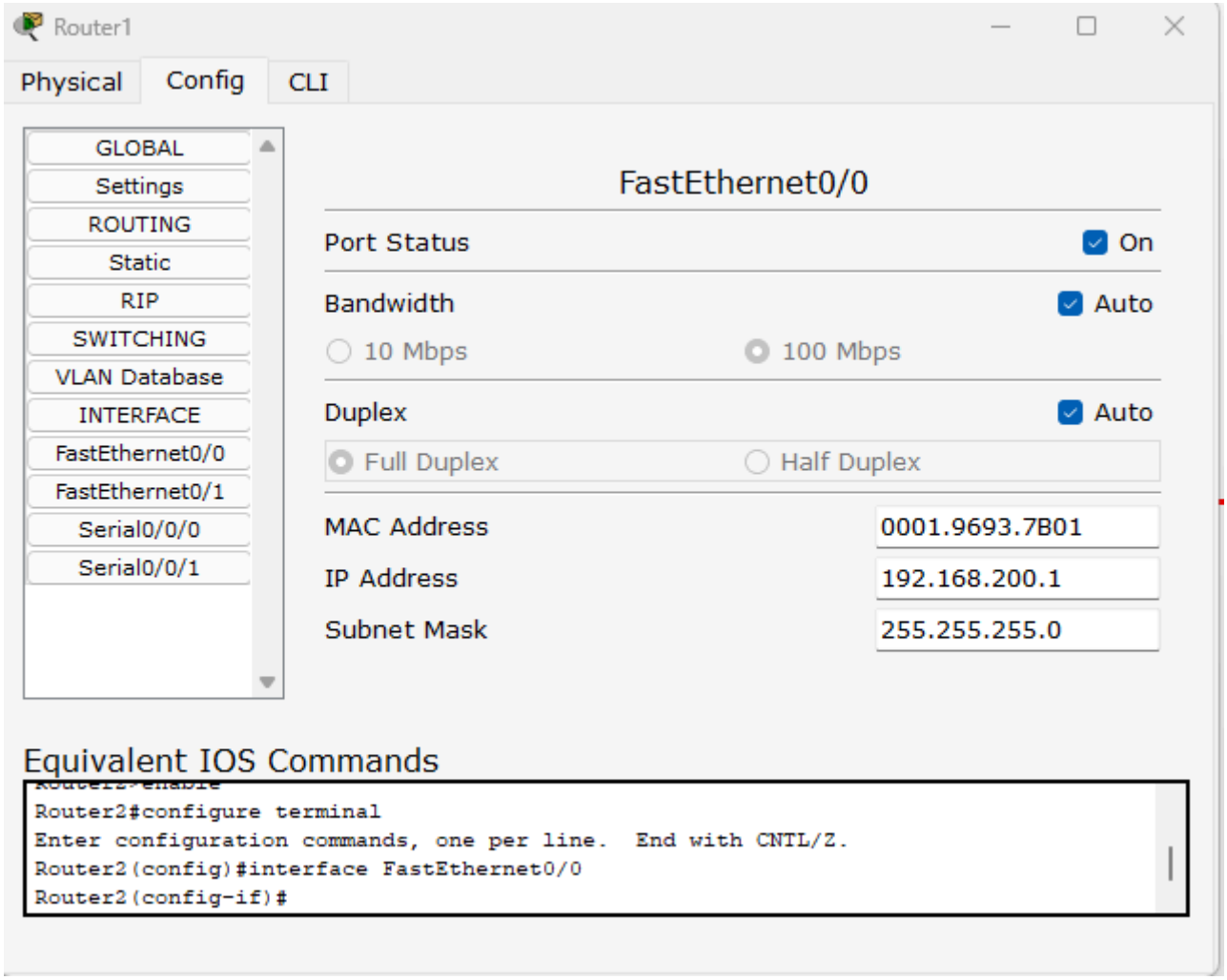



**FIG 10**

Step 7: Try to send Packets Through-out the network

At first attempt it will failed (to share outside the network) but in second attempt of sending packets it will success as shown in **FIG8**

| Fire | Last Status | Source | Destination | Type | Color | Time (sec) | Periodic | Num | Edit | Delet |
|---|---|---|---|---|---|---|---|---|---|---|
| | Failed | Server | PC3 | ICMP | | 0.000 | N | 0 | (edit) | (delet |
| | Successful | Server | PC3 | ICMP | | 0.000 | N | 1 | (edit) | (delet |

**FIG 8**